

Exp 3 Program

```
clc;
```

```
clear;
```

```
close all;
```

```
% Parameters Am = 2; % Message
```

```
Amplitude (V) fm = 200; % Message
```

```
Frequency (Hz) Ac = 4; % Carrier
```

```
Amplitude (V) fc = 2000; % Carrier
```

```
Frequency (Hz) fs = 20*fc; %
```

```
Sampling Frequency (Hz) kp = pi/4;
```

```
% Phase Sensitivity (rad/V)
```

```
% Time Vector
```

```
t = 0:1/fs:4/fm;
```

```
% Message Signal m =
```

```
Am*sin(2*pi*fm*t);
```

```
% Carrier Signal c =
```

```
Ac*cos(2*pi*fc*t);
```

```
% Phase Modulated Signal pm =
```

```
Ac*cos(2*pi*fc*t + kp*m);
```

```
% Phase Deviation phase_dev
```

```
= kp*Am; % PM Modulation
```

```
Index mp = phase_dev;
```

```
% Display Results fprintf('Message Amplitude = %.2f
```

```
V\n',Am); fprintf('Message Frequency = %.0f
```

```
Hz\n',fm); fprintf('Carrier Amplitude = %.2f V\n',Ac);
```

```
fprintf('Carrier Frequency = %.0f Hz\n',fc);
```

```
fprintf('Phase Sensitivity = %.2f rad/V\n',kp);
```

```
fprintf('Phase Deviation = %.2f radians\n',phase_dev);
```

```
fprintf('PM Modulation Index = %.2f\n',mp);
```

```
% Plotting figure('Name','Phase Modulation and Phase  
Deviation',...
```

```
'NumberTitle','off');
```

```
subplot(2,2,1);
```

```
plot(t,m,'LineWidth',1.5);
```

```
title('Message Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude'); grid
```

```
on;
```

```
subplot(2,2,2);
```

```
plot(t,c,'LineWidth',1.5);
```

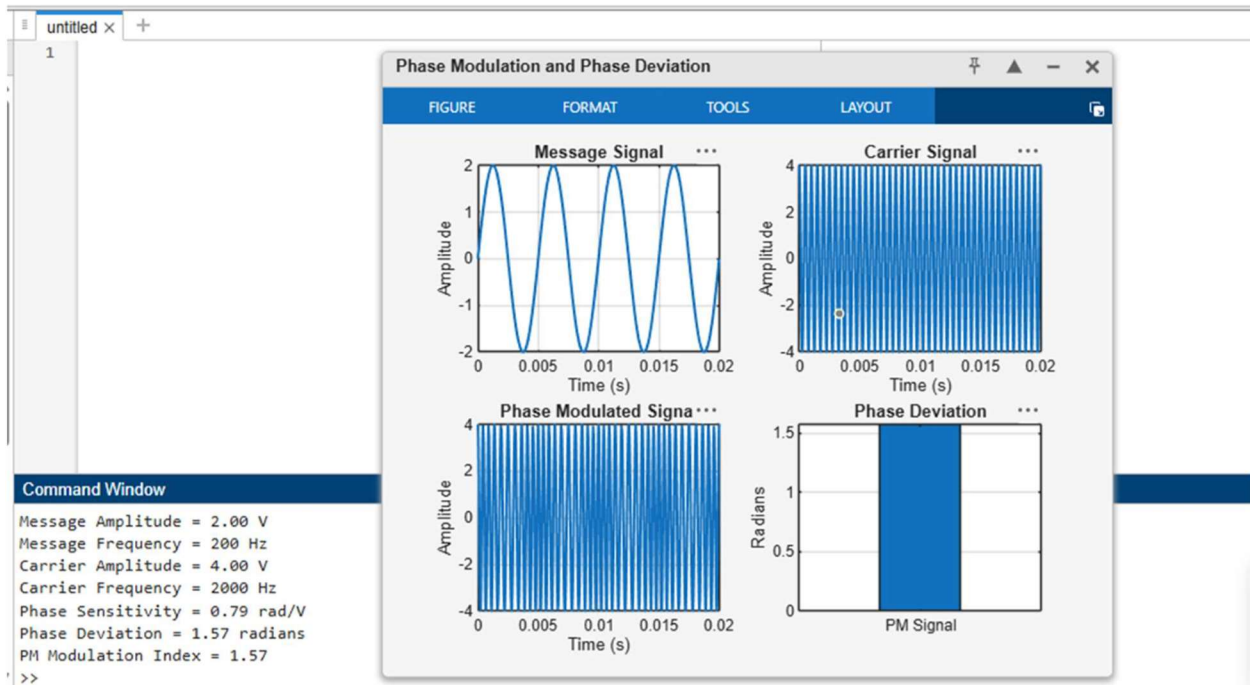
```
title('Carrier Signal');
```

```
xlabel('Time (s)');  
ylabel('Amplitude');  
grid on;
```

```
subplot(2,2,3);  
plot(t,pm,'LineWidth',1.5);  
title('Phase Modulated Signal');  
xlabel('Time (s)');  
ylabel('Amplitude'); grid on;
```

```
subplot(2,2,4); bar(phase_dev);  
title('Phase Deviation');  
ylabel('Radians');  
set(gca,'XTickLabel',{'PM  
Signal'}); grid on;
```

Result:



Exp 3b

Program

clc;

clear;

close all;

% Parameters $A_m = 2$; % Message

Amplitude (V) $f_m = 200$; % Message

Frequency (Hz) $A_c = 4$; % Carrier

Amplitude (V) $f_c = 2000$; % Carrier

Frequency (Hz) $f_s = 20 \cdot f_c$; %

Sampling Frequency (Hz) $k_p = \pi/4$;

% Phase Sensitivity (rad/V)

% Time Vector

$t = 0:1/f_s:4/f_m$;

% Message Signal $m =$

$A_m \cdot \sin(2 \cdot \pi \cdot f_m \cdot t)$;

% PM Signal $p_m =$

$A_c \cdot \cos(2 \cdot \pi \cdot f_c \cdot t + k_p \cdot m)$;

% Phase Deviation β

$= k_p \cdot A_m$;

```
% FFT
```

```
N = length(pm);
```

```
f = linspace(-fs/2,fs/2,N);
```

```
PM_FFT = abs(fftshift(fft(pm)))/N;
```

```
% Power Spectral Density
```

```
PSD = (abs(fftshift(fft(pm))).^2)/(N*fs);
```

```
% Approximate Bandwidth
```

```
BW = 2*(beta+1)*fm;
```

```
fprintf('Phase Deviation = %.2f radians\n',beta); fprintf('Estimated  
Bandwidth = %.2f Hz\n',BW);
```

```
% Plotting figure('Name','PSD Analysis of PM  
Signal','NumberTitle','off');
```

```
subplot(2,2,1);
```

```
plot(t,m); title('Message  
Signal'); xlabel('Time  
(s)');
```

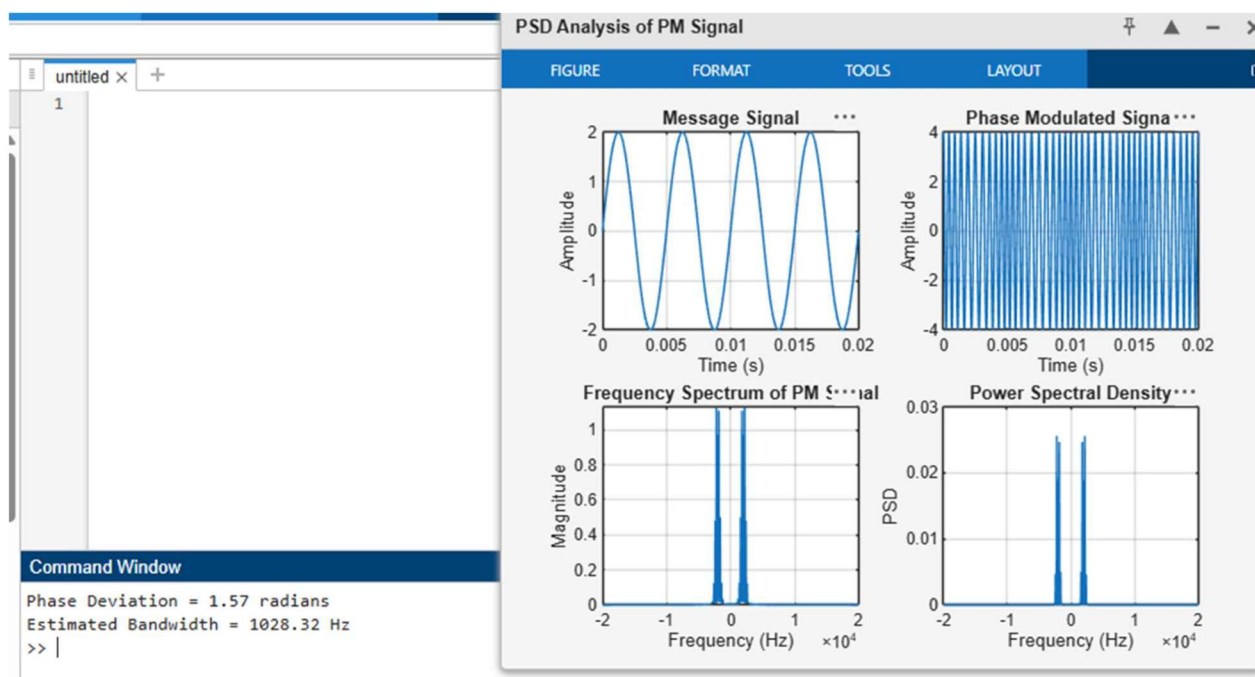
```
ylabel('Amplitude');
```

```
grid on;
```

```
subplot(2,2,2); plot(t,pm);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude'); grid on;
```

```
subplot(2,2,3);
plot(f,PM_FFT); title('Frequency
Spectrum of PM Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude'); grid on;
```

```
subplot(2,2,4); plot(f,PSD);
title('Power Spectral Density');
xlabel('Frequency (Hz)');
ylabel('PSD'); grid on; Result:
```



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